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COURSE PACK

## Syllabus

# Incident Response Fundamentals

An 8-week course from reading one EDR alert to running a ransomware incident through every NIST 800-61 phase.

| | |
|---|---|
| **Course code** | [DEPT 000] |
| **Term** | [Term and year] |
| **Instructor** | [Name] |
| **Contact** | [Email] |
| **Office hours** | [Days, times, location or link] |
| **Meeting pattern** | [Days, times, room or lab] |
| **Length** | 8 weeks, one lab per week |
| **Discipline** | Incident Response |
| **Level** | Upper undergraduate, graduate, or professional cohort |
| **Prerequisites** | Networking and operating systems fundamentals. Introduction to Digital Forensics is helpful but not required. |
| **Lab time** | About 6 hours of hands-on work across the term |
| **Prepared** | 2026-10-18 |

**Bracketed fields are for the adopting instructor to complete. Everything else can be used as written. Every lab in this course generates a different variant of its case for each student, with an answer key verified against that student's own evidence; grading and the student portal run on the instructor's machine.**

# 1. Course description

This course trains the responder's first week on the job: reading detections, deciding what is real, scoping an intrusion and choosing what to do about it. It opens on single sources (an EDR alert, a SIEM queue, proxy logs) and builds to full incidents worked under time. Two 45-minute drills and a 90-minute final incident are run as timed exams in the student portal with integrity logging, because responders work on a clock. Every case is mapped to MITRE ATT&CK and the NIST 800-61 lifecycle, and every student works a different variant, so the class can be debriefed on decisions rather than on answers.

# 2. Learning outcomes

On completing the course a student can:

1. Triage EDR, SIEM and proxy evidence by severity and ATT&CK technique.
2. Scope an intrusion from initial access to impact.
3. Choose and justify containment, eradication and recovery actions.
4. Produce indicators and a breach report a manager can act on.

# 3. Prerequisites and materials

## Prerequisites

Networking and operating systems fundamentals. Introduction to Digital Forensics is helpful but not required.

## Software

All tools are free. One-run install scripts for each are on the Tools page of the product website.

| Tool | Used for |
|---|---|
| Any JSON viewer | EDR and SIEM exports |
| jq | filtering large exports on the command line |
| Excel or LibreOffice | sorting and pivoting alert and proxy tables |
| Wireshark | packet captures |
| EvtxECmd | Windows event logs to CSV |
| Timeline Explorer | reading the CSVs and building timelines |

## Hardware

A laptop or lab workstation with 8 GB of RAM (16 GB for memory forensics weeks), 20 GB of free disk and permission to install the listed tools. Windows, macOS and Linux all work; some tools are Windows-only and run in a virtual machine elsewhere.

## Evidence

Evidence is generated per student and handed out as a ZIP or through a private portal link each week. No textbook purchase is required; each week names its reading from the free references below.

### Readings and references

No purchase is required. The texts below are free from their publishers, and each week names the part to read. Editions and addresses change; the title and publisher are the stable reference.

| Title | Publisher | Use in this course |
|---|---|---|
| Computer Security Incident Handling Guide (SP 800-61 Rev. 2; Rev. 3 is the CSF 2.0 profile) https://doi.org/10.6028/NIST.SP.800-61r2 | NIST | The lifecycle every week maps to: preparation, detection and analysis, containment, eradication and recovery, lessons learned. |
| MITRE ATT&CK Enterprise matrix and Getting Started https://attack.mitre.org/ | MITRE | Technique ids for every detection from week 1; tactics for scoping in weeks 6 and 8. |
| Federal Government Cybersecurity Incident and Vulnerability Response Playbooks https://www.cisa.gov/resources-tools/resources/federal-government-cybersecurity-incident-and-vulnerability-response-playbooks | CISA | Decision checklists for the drills in weeks 4 and 7. |
| Guide to Malware Incident Prevention and Handling (SP 800-83 Rev. 1) https://doi.org/10.6028/NIST.SP.800-83r1 | NIST | Malware triage and containment, week 5. |
| #StopRansomware Guide https://www.cisa.gov/stopransomware/ransomware-guide | CISA and partners | Response checklist for the final incident, week 8. |
| Windows security auditing: events to monitor https://learn.microsoft.com/windows-server/identity/ad-ds/plan/appendix-l--events-to-monitor | Microsoft Learn | Logon types and service installs for lateral movement, week 6. |
| Responding to a compromised email account https://learn.microsoft.com/defender-office-365/responding-to-a-compromised-email-account | Microsoft Learn (Defender for Office 365) | Mailbox rules, sign-ins and what to revoke, week 7. |
| Wireshark User's Guide https://www.wireshark.org/docs/wsug_html_chunked/ | Wireshark Foundation | Reading captures behind the proxy and C2 weeks. |
| Good Practice Guide for Incident Management https://www.enisa.europa.eu/publications/good-practice-guide-for-incident-management | ENISA | Team roles and escalation; background for the breach report. |

## 4. How a week runs

1. Each week has one lab built on a single case. Every student receives a different variant of that case: different names, hosts, timestamps and answers, generated from the same story.

2. Before the lab, read the briefing and the week's preparation notes, and confirm the listed tools open on your machine.

3. During the lab, work the evidence with the listed tools and record each answer with the artifact and method that proves it.

4. Submit through the student portal link for that week, or hand in the worksheet. Portal submissions are graded against your own answer key the moment you submit.

5. After release, review the walkthrough for your variant: where each answer was, and how it was found.

# 5. Assessment and grading

| Component | Weight | Details |
|---|---|---|
| Weekly labs (5) | 50% | Best 4 of 5 count. Auto-graded on submission; each lab is 10 to 14 questions on that week's case. |
| Timed drills (2) | 30% | Weeks 4 and 7. Two 45-minute exams in the portal, 15% each, each ending in a containment or revocation decision. |
| Final incident | 20% | Week 8. A 90-minute timed exam on a live ransomware chain, plus a breach report due one week later. |

## Grading scale

| | |
|---|---|
| A | 90 to 100 |
| B | 80 to 89 |
| C | 70 to 79 |
| D | 60 to 69 |
| F | below 60 |

Timed assessments are graded on submission in the portal. Reports are graded with the rubric in section 8. Scores export to the institution's LMS from each week's job.

# 6. Course policies

## Collaboration

Discussing tools and technique with classmates is encouraged. Answers are not transferable: each variant has its own evidence and its own key, and a submission that matches another student's variant rather than your own is flagged by the collusion report and treated as academic misconduct.

## Late work

Labs are due at the end of the week they are set. Late labs lose 10% per day for three days and are not accepted after that. The lowest lab scores are dropped as stated in the assessment table, which covers illness and emergencies without paperwork for short absences.

## Timed assessments

Timed exams run in the student portal with a fixed clock, shuffled question order and integrity logging (tab switches, full-screen exits and pastes are recorded). Arrive with your tools installed and tested.

## Resubmission

One lab per term may be resubmitted within a week of feedback for up to 80% of full credit.

## Evidence handling

Evidence files are synthetic. Treat them as you would real evidence anyway: verify hashes, work on copies, and document every step. That habit is assessed.

## Accommodations

[Insert the institution's disability accommodation and extended-time statement. The portal supports per-student time extensions.]

## Academic integrity

[Insert the institution's academic integrity statement.]

## 7. Schedule

One case per week. The mode and time budget are set when the course is generated; the deliverable is what the student hands in.

| Wk | Topic | Case | Mode | Assessment |
|---|---|---|---|---|
| 1 | Reading an EDR alert | First Look: Reading an EDR Alert | Practice, about 25 min | Lab 1 |
| 2 | The SIEM queue | First Look: Reading a SIEM Alert | Practice, about 25 min | Lab 2 |
| 3 | Beacons in web traffic | First Look: One Beacon in the Web Traffic | Practice, about 30 min | Lab 3 |
| 4 | Phishing to foothold | Phishing to Foothold | Timed exam, 45 min | Drill 1 (15%) |
| 5 | Malware triage | Malware Triage: From Alert to Verdict | Practice, about 35 min | Lab 4 |
| 6 | Lateral movement | Lateral Move to the File Server | Practice, about 60 min | Lab 5 |
| 7 | Business email compromise | Business Email Compromise: Wire Fraud | Timed exam, 45 min | Drill 2 (15%) |
| 8 | Final incident: ransomware | Ransomware Before Coffee | Timed exam, 90 min, with report | Final 20% |

# Week by week

Each week below lists its objectives, the case and what it contains, what students should do beforehand, what they hand in, and a note for the instructor.

## Week 1: Reading an EDR alert

### Objectives

• Triage by severity and read a process tree.

• Map a detection to an ATT&CK technique.

### Case

First Look: Reading an EDR Alert. Your first shift on the SOC queue. One endpoint threw a short stack of EDR detections. Read the feed the way a responder does - triage by severity, read the process tree and the ATT&CK technique on each one - and decide whether this is a real incident or noise.

Practice, about 25 min · 12 questions · Tools: Any JSON viewer, jq, Excel

### Reading

SP 800-61r2 section 3.2 (detection and analysis). ATT&CK: Getting Started and the Execution tactic.

### Before the lab

Read the NIST 800-61 incident lifecycle summary and the ATT&CK matrix overview.

### Deliverable

Lab 1 answers, each with the ATT&CK technique id.

### Instructor note

Insist on technique ids from day one; the rest of the course assumes the vocabulary.

## Week 2: The SIEM queue

### Objectives

• Read a SIEM alert export.

• Separate a true positive from noise.

### Case

First Look: Reading a SIEM Alert. Your first shift on the SOC queue. One correlated SIEM export is waiting with a short stack of alerts. Read it the way a responder does - sort by severity, open the worst alert, and pull its rule, technique and the source, destination and user entities so you can escalate cleanly.

Practice, about 25 min · 12 questions · Tools: Any JSON viewer, jq, Excel

### Reading

SP 800-61r2 section 3.2.4 (incident analysis) and 3.2.6 (prioritisation).

### Before the lab

Open the sample SIEM export in a JSON viewer and in a spreadsheet; decide which you prefer.

### Deliverable

Lab 2 answers and a one-line triage note per alert: true positive, benign, or needs more data.

### Instructor note

Most of the queue is noise by design; grade the dismissals as carefully as the escalation.

## Week 3: Beacons in web traffic

### Objectives

• Spot periodic egress in proxy logs.

• Pull a C2 indicator set.

### Case

First Look: One Beacon in the Web Traffic. A workstation's outbound traffic for one hour was captured off the sensor. Most of it is ordinary browsing to normal sites. Buried in it, the host looks up a domain nobody recognizes, then calls back to it again and again on a schedule, and finally pushes a batch of data out to it. Read the capture the way a responder reads a proxy log - rule out the normal destinations, find the one that beacons, and pull the domain, IP, cadence and the size of what left.

Practice, about 30 min · 10 questions · Tools: Wireshark, tshark, NetworkMiner

### Reading

ATT&CK: Command and Control tactic, T1071 and T1573. Wireshark User's Guide chapter 6.

### Before the lab

Read the beaconing primer: periodicity, jitter, user-agent anomalies.

### Deliverable

Lab 3 answers and an indicator list (domains, IPs, user agents) ready to block.

### Instructor note

Have students compute the beacon interval by hand once before letting a tool do it.

## Week 4: Phishing to foothold

### Objectives

• Follow a phish through delivery, execution and persistence.

• Make the first containment call under time.

### Case

Phishing to Foothold. A user clicked. Walk the intrusion end to end: read the phishing email (sender, originating IP, subject and attachment), follow the payload as it executes on the endpoint, and pull the one C2 indicator that ties it together. The domain the mail came from is the domain the payload beacons to - find it and the team can block both at once.

Timed exam, 45 min · 14 questions · Tools: Thunderbird, Any JSON viewer, jq, Excel

### Reading

SP 800-61r2 section 3.3 (containment). CISA playbook: incident response, containment step.

### Before the lab

Review weeks 1 to 3. Read the containment decision checklist.

### Deliverable

Drill 1 submitted in the portal within 45 minutes, including the containment call.

### Instructor note

Exam mode. Debrief the containment decisions as a group; there is more than one defensible answer.

## Week 5: Malware triage

### Objectives

• Go from alert to verdict with hashes, signatures and behaviour.

### Case

Malware Triage: From Alert to Verdict. A single endpoint threw a short stack of EDR detections. Work it from alert to verdict: find the worst severity and the technique, read what the sensor actually did, pull the offending process, its hash and parent, pin the first-detection time, and set the incident severity.

Practice, about 35 min · 14 questions · Tools: Any JSON viewer, jq, Excel, Timeline Explorer

### Reading

SP 800-83r1 sections 3 and 4 (detection, containment, eradication of malware).

### Before the lab

Read the malware triage primer: hashes, signatures, sandbox behaviour, when to stop.

### Deliverable

Lab 4 answers and a verdict with severity.

### Instructor note

The case rewards stopping at a verdict; students who try to reverse the sample run out of time.

## Week 6: Lateral movement

### Objectives

• Pair 4624 logons with 7045 service installs.

• Scope which hosts the attacker reached.

### Case

Lateral Move to the File Server. A foothold on one workstation becomes access to the file server. Using the Security and System logs, the registry and Prefetch, trace the remote-service execution: where the session came from, the service that ran the attacker's tool, the command it launched, and the Run key left to survive a reboot.

Practice, about 60 min · 13 questions · Tools: Event Viewer, EvtxECmd, Registry Explorer, PECmd

### Reading

Microsoft events to monitor: 4624 logon types, 4648, 7045, 4698. ATT&CK: Lateral Movement, T1021.

### Before the lab

Install EvtxECmd and Timeline Explorer. Read the lateral movement primer: 4624 logon types, 7045, 4698.

### Deliverable

Lab 5 answers and a scoping table of hosts reached, with the evidence for each.

### Instructor note

The scoping table is the artefact a real responder hands to leadership; grade it as such.

## Week 7: Business email compromise

### Objectives

• Trace a fraudulent wire request through mail and sign-in evidence.

• Decide what to revoke first.

### Case

Business Email Compromise: Wire Fraud. A finance controller wired money to an attacker's account after a spoofed executive email and a lookalike vendor followed up with new bank details.

Timed exam, 45 min · 13 questions · Tools: Thunderbird, Autopsy, DB Browser for SQLite

### Reading

Microsoft: responding to a compromised email account. ATT&CK: T1114 and T1564.008.

### Before the lab

Read the BEC primer: mailbox rules, OAuth consents, impossible travel.

### Deliverable

Drill 2 submitted within 45 minutes, including what to revoke first and why.

### Instructor note

Exam mode. Time pressure is realistic here; a wire is pending.

## Week 8: Final incident: ransomware

### Objectives

• Run a live ransomware chain through detection, containment, eradication and recovery.

• Write the breach report.

### Case

Ransomware Before Coffee. You are the on-call responder. The EDR console lit up at 06:48 with a chain of detections on a finance workstation, ending in a running locker. Triage the detection feed and the host's own event logs, scope the intrusion from first access to impact, pull the indicators, map it to ATT&CK, and make the containment call - fast.

Timed exam, 90 min, with report · 15 questions · Tools: Any JSON viewer, jq, Excel, Event Viewer, EvtxECmd, Timeline Explorer

### Reading

SP 800-61r2 sections 3.3 to 3.4 and the #StopRansomware Guide response checklist. ENISA guide on reporting.

### Before the lab

Review every week. Read the breach report template and rubric.

### Deliverable

Final incident submitted within 90 minutes; breach report due one week later.

### Instructor note

Run with the report question set. The report is the deliverable a manager would read; grade it on decisions and evidence, not narrative.

## 8. Report rubric

Applied to the written report that follows the final timed assessment. The report is written from the student's own saved answers for their own variant, so two reports in the same class never describe the same facts.

| Criterion | Weight | Full credit looks like |
|---|---|---|
| Scope | 30% | Initial access, every host and account reached, and the data touched, each with the evidence that proves it. |
| Decisions | 30% | Containment, eradication and recovery actions are specific, ordered, and justified against the evidence. |
| Indicators | 20% | A usable indicator list: hashes, domains, IPs, accounts, with where each was seen. |
| Timeline | 10% | One normalised timezone, in order, mapped to ATT&CK and NIST phases. |
| Writing | 10% | A manager can act on page one; detail follows; no speculation presented as fact. |

## 9. Running the course in Red Herrings

1. Open Courses, pick this course, enter the class size or import the roster CSV, and press Generate the whole course. One job per week is created with the mode and time limit in the schedule.

2. Each week, open that week's job and hand out the student ZIPs or the private portal links. Portal links can be exported as a CSV for the LMS.

3. Timed weeks run in exam mode with integrity logging. Release scores from the job page after the sitting.

4. Grade from the job page or the Grading screen; export to Canvas, Blackboard, Moodle, D2L or QTI, or let LTI pass grades back.

5. Term Analytics shows the whole course at the end of term: scores by week, the questions most often missed, and integrity events.

Syllabus text, policies and the rubric may be copied into institutional templates without attribution.